

4.3 Exercises

5 Corrective Maintenance (CM) Work

5.1 CM Work Overview

Corrective maintenance (CM) work is a maintenance work initiated based on failure happens on assets or locations. The purpose of this maintenance work is to return an asset or location into a satisfactory operating condition. CM work order created in **Work Order Tracking** application.

Planning should be held before executing a CM work. A job plan can be defined in a CM work order to automate the planning resources and tasks need to be performed, or creator can defined them manually. Approval from the CM work order is needed when there are materials required for the work.

Most CM work requires downtime of the assets to be performed, but not always. When downtime required, the work schedule need to be considered more carefully, to minimize the impact on business process. If the schedule does not fit, a schedule revision should be done towards those CM work order.

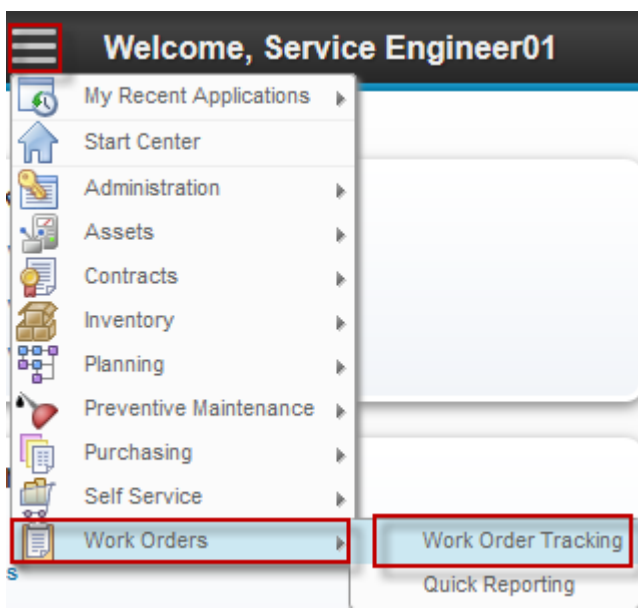
5.2 CM Work Cycle

For the details of corrective maintenance process flow, please refers to below image.

3. When there are materials planned to perform this CM work, an approval assignment will be sent to *CHE Engineer* to issue those materials from the storeroom. If the materials are unavailable in the storeroom, procurement should be held to purchase them.
4. *Operation* approval is required related to CM work order schedule, if the work order requires downtime. *Service Engineer* should revise the schedule if it does not fit.
5. *Service Engineer* reviews the work result and input the result into EMS. A closing approval assignment will be sent to *CHE Engineer*, whether the work order can be closed or not.

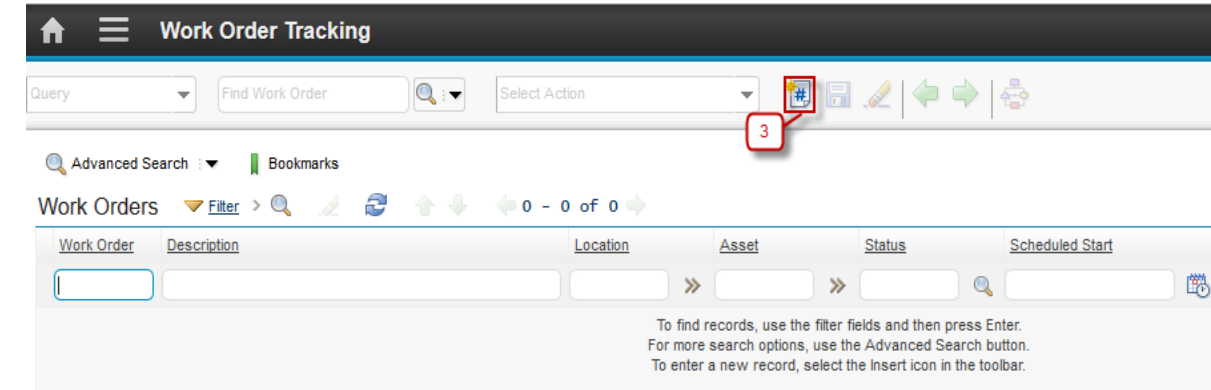
5.2.1 CM Work Creation & Submission

Below are step by step for CM Work Order creation & submission:

Step	Action
1. Login as Service Engineer. User name : se01, password : password.	
2. Go to Work Order Application by click  (GoTo) – Work Orders – Work Order Tracking	
3. Click New Work Order  button 3a. Work Order number is autonumber. 3b. Input the Work Order description. 3c. Select Work Type 'CM' (Corrective Maintenance) . 3d. Select Asset (mandatory field). Location automatically filled in accordance with the location of assets. 3e. Input Schedule Start and Schedule Finish. 3f. Check the Requires Asset Downtime? 3g. Select the Job Plan.	

3h. Click  **Save** button

3i. The **Route Workflow** button appearance change to , indicates that the workflow is started.



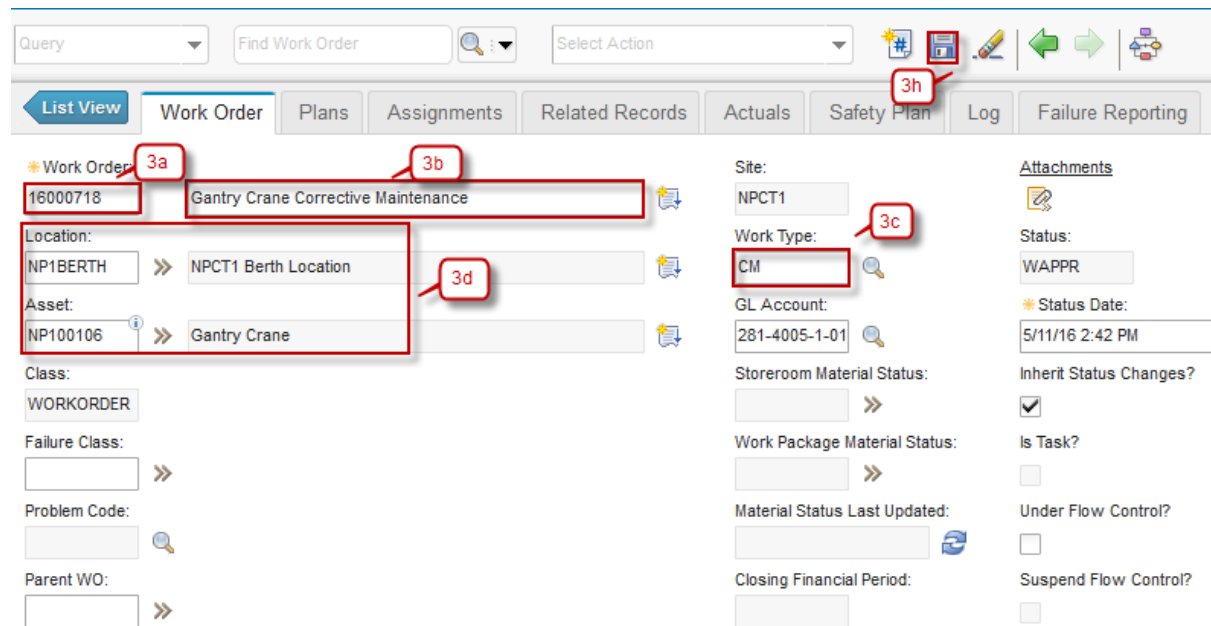
Work Order Tracking

Query Find Work Order Select Action

Advanced Search Bookmarks

Work Orders Filter 0 - 0 of 0

To find records, use the filter fields and then press Enter.
For more search options, use the Advanced Search button.
To enter a new record, select the Insert icon in the toolbar.



Query Find Work Order Select Action

List View Work Order Plans Assignments Related Records Actuals Safety Plan Log Failure Reporting

Work Order 3a

16000718 3b Gantry Crane Corrective Maintenance

Location: 3d
NP1BERTH >> NPCT1 Berth Location

Asset:
NP100106 >> Gantry Crane

Class:
WORKORDER

Failure Class:
>>

Problem Code:
>>

Parent WO:
>>

Site:
NPCT1

Work Type: 3c
CM

GL Account:
281-4005-1-01

Storeroom Material Status:
>>

Work Package Material Status:
>>

Material Status Last Updated:
>>

Closing Financial Period:
>>

Attachments

Status:
WAPPR

Status Date:
5/11/16 2:42 PM

Inherit Status Changes?
☒

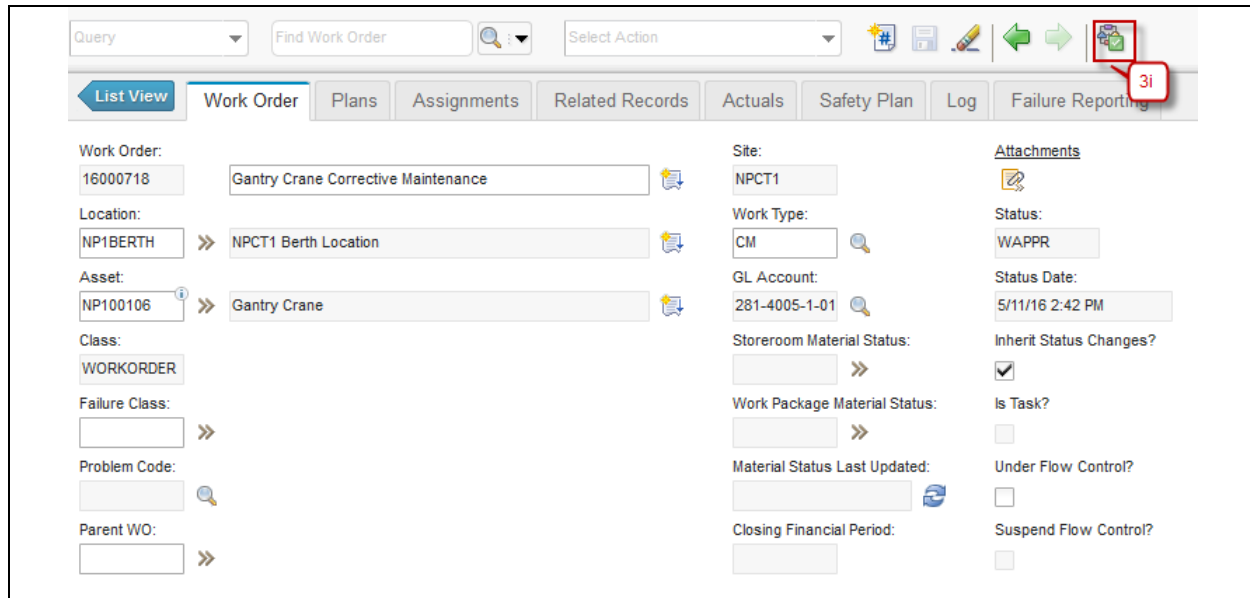
Is Task?
☐

Under Flow Control?
☐

Suspend Flow Control?
☐

Scheduling Information		Follow-up Work
Target Start:	Actual Start:	Is Rework?
		☐
Target Finish:	Actual Finish:	Originating Record:
		>>
Scheduled Start:	* Duration:	Has Follow-up Work?
5/11/16 2:44 PM	0:00	☐
Scheduled Finish:	Predecessors:	
5/12/16 2:42 PM	>>	
Start No Earlier Than:	Include Tasks in Schedule?	
	☒	
Finish No Later Than:		
Requires Asset Downtime?		
☒		

Job Details	Asset Details	Priority
Job Plan:	Asset Up?	Priority:
JP01 >>	☒	
Job Plan Revision #:	SLA Applied?	Priority Justification:
0	☐	
PM:	Charge to Store?	
>>	☐	
Safety Plan:		
>>		



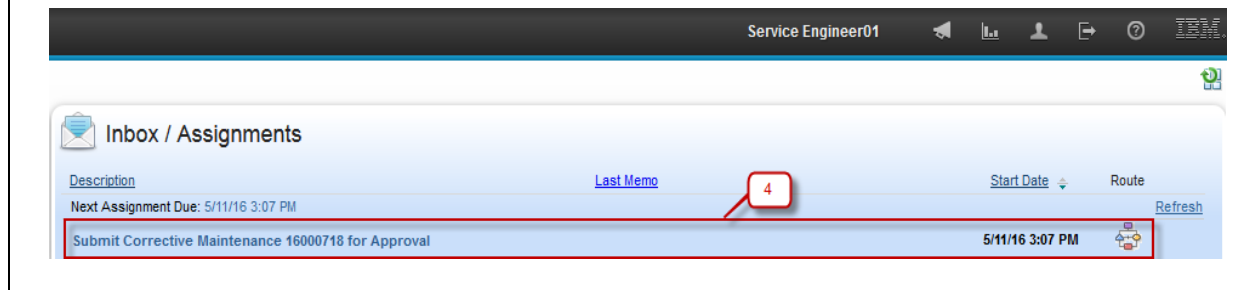
4. Go to Start Center Service Engineer, select work order assignment that displayed on **Inbox/Assignment** column.

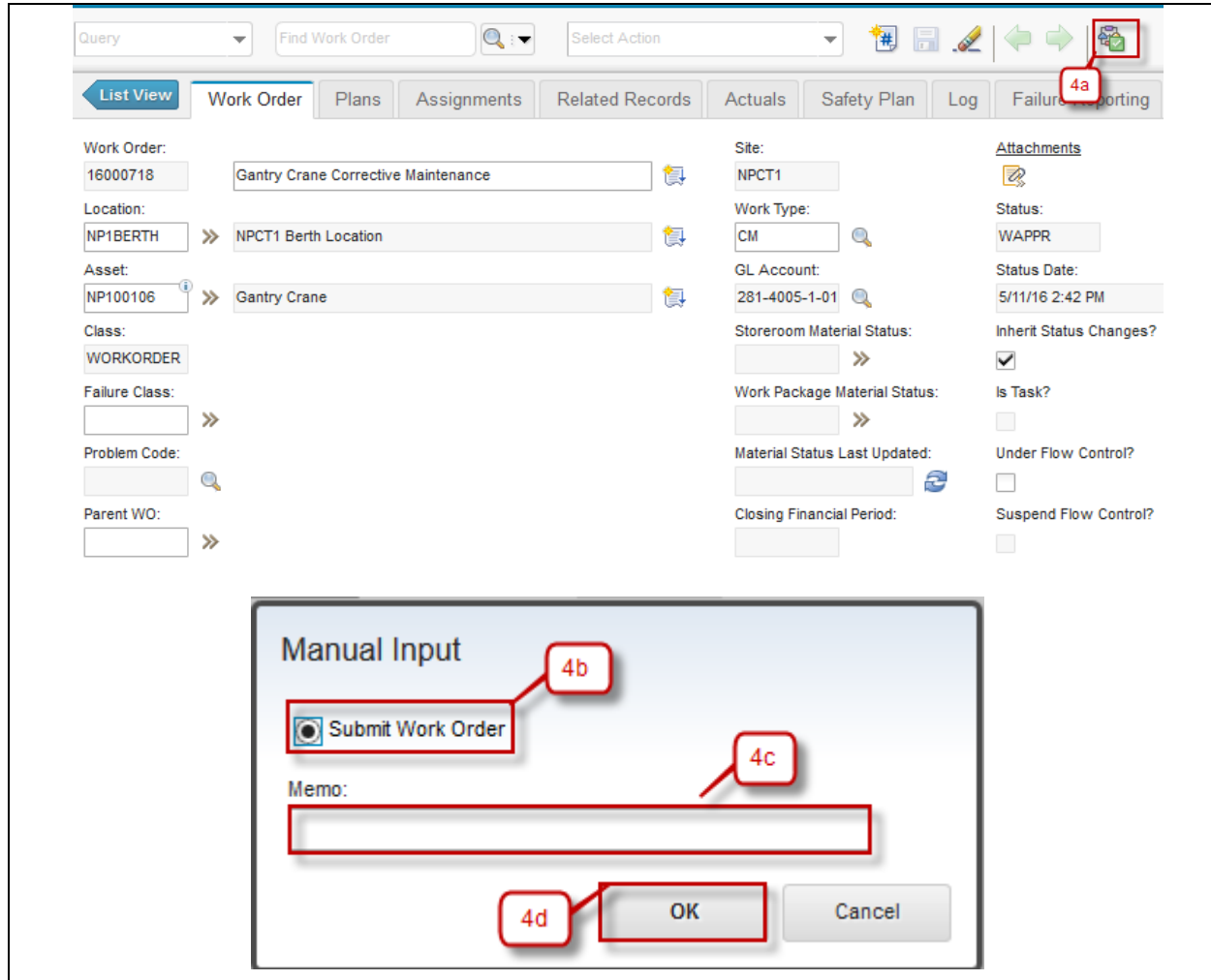
4a. Click  (Route Workflow) button and dialog box to submit Work Order will appears.

4b. Choose **Submit Work Order**.

4c. Input Memo (if needed).

4d. Click **OK** button.





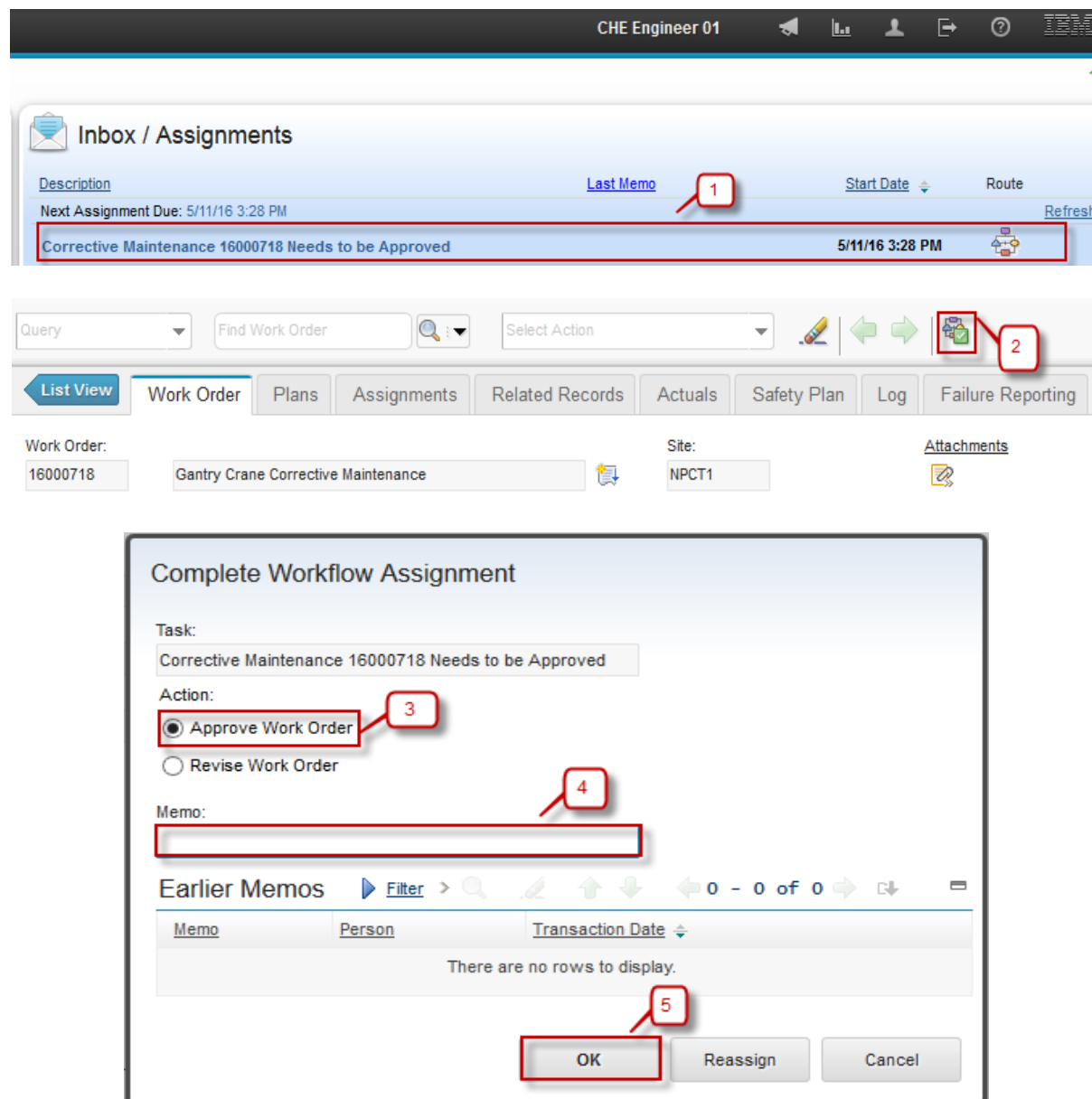
The screenshot displays the Agile Work Management interface. At the top, there is a navigation bar with tabs: List View, Work Order (selected), Plans, Assignments, Related Records, Actuals, Safety Plan, Log, and Failure Reporting. A red box labeled '4a' highlights the 'Failure Reporting' tab. Below the navigation bar, the 'Work Order' form is visible. It contains fields for Work Order (16000718), Location (NP1BERTH), Asset (NP100106), Class (WORKORDER), Failure Class, Problem Code, Parent WO, Site (NPCT1), Work Type (CM), GL Account (281-4005-1-01), Storeroom Material Status, Work Package Material Status, Material Status Last Updated, Closing Financial Period, Attachments, Status (WAPPR), Status Date (5/11/16 2:42 PM), Inherit Status Changes? (checked), Is Task? (unchecked), Under Flow Control? (unchecked), and Suspend Flow Control? (unchecked). A red box labeled '4b' highlights the 'Submit Work Order' button in the 'Manual Input' dialog box. A red box labeled '4c' highlights the 'Memo' text area. A red box labeled '4d' highlights the 'OK' button.

5.2.2 CM Work Plan Approval Process by CHE Engineer

CHE Engineer approval is required when material is planned for the work order. Below are steps to approve CM work plan by *CHE Engineer*. For rejection, revision request, and revision will not be discussed. They can refer to rejection, revision request, and revision of **SM Work Cycle** in step point **1.2.5 SM Request Revision Work Order** and **1.2.6 SM Revise and Re-submit Work Order** training material.

As a note, request revision for CM work plan by *CHE Engineer* will direct the assignment back to *Service Engineer*. After revision is done, assignment will be directed back to *CHE Engineer* for plan approval.

- | Step | Action |
|------|---|
| 5. | Login as CHE Engineer. Username : che01, password: password. On Start Center CHE Engineer an assignment to approve created Work Order will appears in Inbox/Assignment portlet, click Description of the assignment to go to related Work Order. |
| 6. | Then click  (route workflow) and dialog box Complete Workflow Assignment will appears. |
| 7. | Choose Approve Work Order . |
| 8. | Input Memo (if needed). |
| 9. | Click OK . |
| 10. | Status change to ' APPR ' (Approve). |



CHE Engineer 01

Inbox / Assignments

Description Last Memo Start Date Route

Next Assignment Due: 5/11/16 3:28 PM Refresh

Corrective Maintenance 16000718 Needs to be Approved 5/11/16 3:28 PM

Query Find Work Order Select Action

List View Work Order Plans Assignments Related Records Actuals Safety Plan Log Failure Reporting

Work Order: 16000718 Gantry Crane Corrective Maintenance Site: NPCT1 Attachments

Complete Workflow Assignment

Task: Corrective Maintenance 16000718 Needs to be Approved

Action:

☒ Approve Work Order ☐ Revise Work Order

Memo:

Earlier Memos Filter 0 - 0 of 0

Memo Person Transaction Date

There are no rows to display.

OK Reassign Cancel

Query
Find Work Order
Select Action

List View
Work Order
Plans
Assignments
Related Records
Actuals
Safety Plan
Log
Failure Reporting

Work Order:
16000718
Gantry Crane Corrective Maintenance

Location:
NP1BERTH
NPCT1 Berth Location

Asset:
NP100106
Gantry Crane

Class:
WORKORDER

Failure Class:

Problem Code:

Parent WO:

Site:
NPCT1

Work Type:
CM

GL Account:
281-4005-1-01

Storeroom Material Status:

Work Package Material Status:
PARTIAL

Material Status Last Updated:
5/11/16 3:45 PM

Closing Financial Period:

Attachments

Status:
APPR

Status Date:
5/11/16 3:45 PM

Inherit Status Changes?

Is Task?

Under Flow Control?

Suspend Flow Control?

5.2.3 CM Work Schedule Approval by Operation

After CM work plan is approved, if asset downtime is required to execute the work, the work schedule needs to be approved by *Operation*. Below are steps to approve a CM work schedule by *Operation*. For rejection, revision request, and revision will not be discussed. They can refer to rejection, revision request, and revision of **SM Work Cycle** in step point **1.2.5 SM Request Revision Work Order** and **1.2.6 SM Revise and Re-submit Work Order** training material.

As a note, request revision for CM work schedule by *Operation* will direct the assignment back to *Service Engineer*. After revision is done, assignment will be directed back to *Operation* for schedule approval.

Step	Action
1.	Login as Operation. Username : ops01, password : password. On Start Center Operation an assignment to review approved Work Order schedule will appears in Inbox/Assignment portlet, click Description of the assignment to go to related Work Order.
2.	Then click  (route workflow) and dialog box Complete Workflow Assignment will appears.
3.	Choose Approve Schedule .
4.	Input Memo (if needed).
5.	Then click OK .
6.	Status change to ' WEXEC ' (Waiting for Execution)

Operation01

Inbox / Assignments

Description

Next Assignment Due: 5/11/16 3:45 PM

Corrective Maintenance 16000718 Schedule Needs to be Reviewed

Last Memo

Route

Refresh

1 - 1 of 1

Query Find Work Order Select Action

List View Work Order Plans Assignments Related Records Actuals Safety Plan Log Failure Reporting

Work Order: 16000718 Gantry Crane Corrective Maintenance Site: NPCT1 Attachments

Complete Workflow Assignment

Task: Corrective Maintenance 16000718 Schedule Needs to be Rev

Action:

☒ Approve Schedule

☐ Revise Schedule

Memo:

Earlier Memos

Filter

0 - 0 of 0

Memo Person Transaction Date

There are no rows to display.

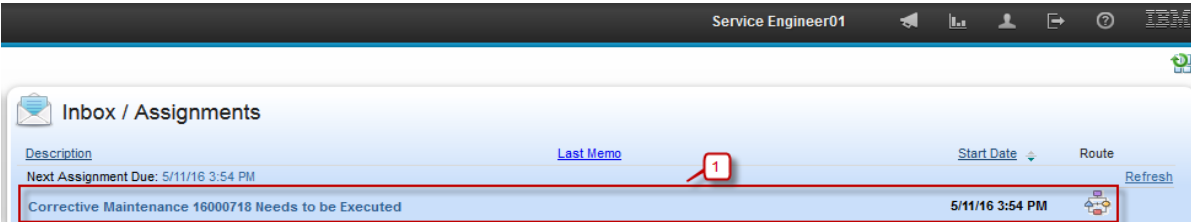
OK Reassign Cancel

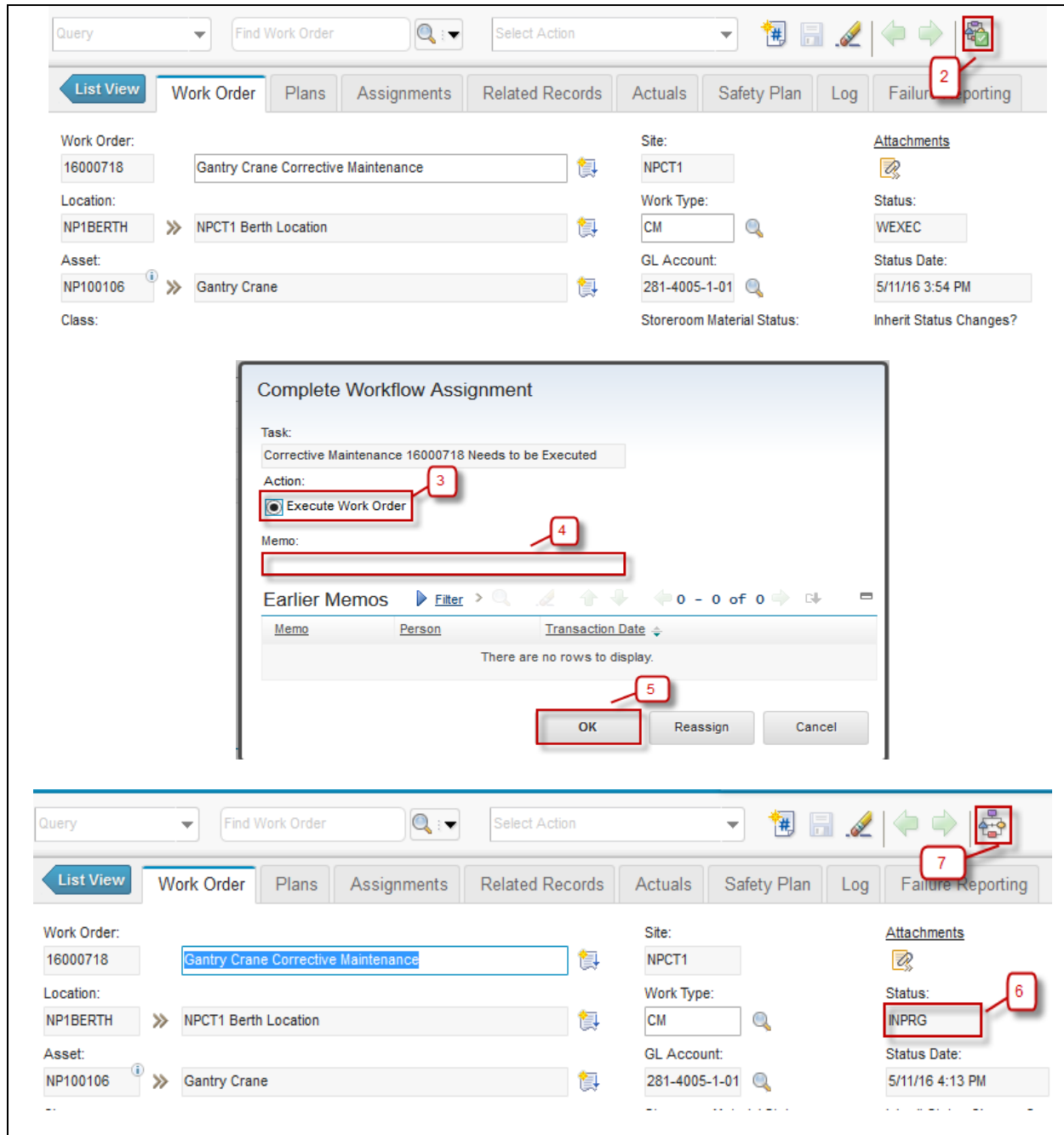
List View	Work Order	Plans	Assignments	Related Records	Actuals	Safety Plan	Log	Failure Reporting
Work Order: 16000718 Gantry Crane Corrective Maintenance		Site: NPCT1		Attachments 				
Location: NP1BERTH NPCT1 Berth Location		Work Type: CM		Status: WEEXEC				
Asset: NP100106 Gantry Crane		GL Account: 281-4005-1-01		Status Date: 5/11/16 3:54 PM				
Class: WORKORDER		Storeroom Material Status: 		Inherit Status Changes? ☒				
Failure Class: 		Work Package Material Status: PARTIAL		Is Task? ☐				
Problem Code: 		Material Status Last Updated: 5/11/16 3:45 PM		Under Flow Control? ☐				
Parent WO: 		Closing Financial Period: 		Suspend Flow Control? ☐				

5.2.4 CM Work Execution

To initiate CM work, below are step by step needs to be done by *Service Engineer* on CM work order:

Step	Action
1.	Login as Service Engineer. Username : se01, password : password. On Start Center Service Engineer an assignment to execute Work Order will appears in Inbox/Assignment portlet, click Description of the assignment to go to related Work Order.
2.	Then click  (route workflow) and dialog box Complete Workflow Assignment will appears.
3.	Choose Execute Work Order .
4.	Input Memo (if needed).
5.	Then click OK .
6.	Status change to 'INPRG' (In Progress).
7.	The Route Workflow button appearance returned back into  , indicates that the workflow is finished.





Query Find Work Order Select Action

List View Work Order Plans Assignments Related Records Actuals Safety Plan Log Failure Reporting

Work Order: 16000718 Gantry Crane Corrective Maintenance

Location: NP1BERTH NPCT1 Berth Location

Asset: NP100106 Gantry Crane

Class:

Site: NPCT1

Work Type: CM

GL Account: 281-4005-1-01

Storeroom Material Status:

Attachments

Status: WEXEC

Status Date: 5/11/16 3:54 PM

Inherit Status Changes?

Complete Workflow Assignment

Task: Corrective Maintenance 16000718 Needs to be Executed

Action: ☒ Execute Work Order

Memo:

Earlier Memos

Memo	Person	Transaction Date
There are no rows to display.		

OK Reassign Cancel

Query Find Work Order Select Action

List View Work Order Plans Assignments Related Records Actuals Safety Plan Log Failure Reporting

Work Order: 16000718 Gantry Crane Corrective Maintenance

Location: NP1BERTH NPCT1 Berth Location

Asset: NP100106 Gantry Crane

Site: NPCT1

Work Type: CM

GL Account: 281-4005-1-01

Attachments

Status: INPRG

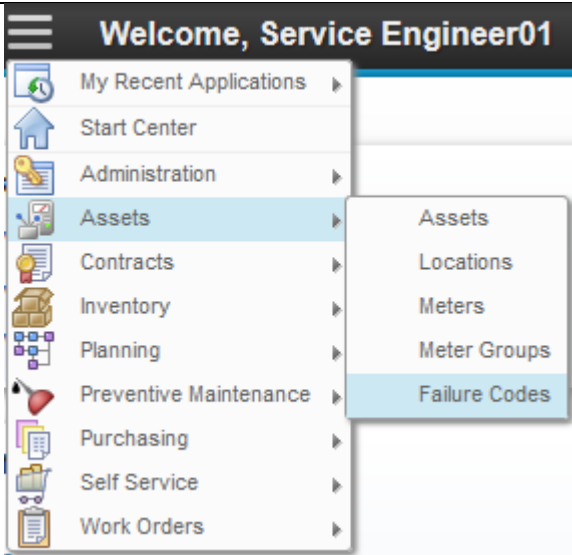
Status Date: 5/11/16 4:13 PM

5.2.5 Failure Codes Application

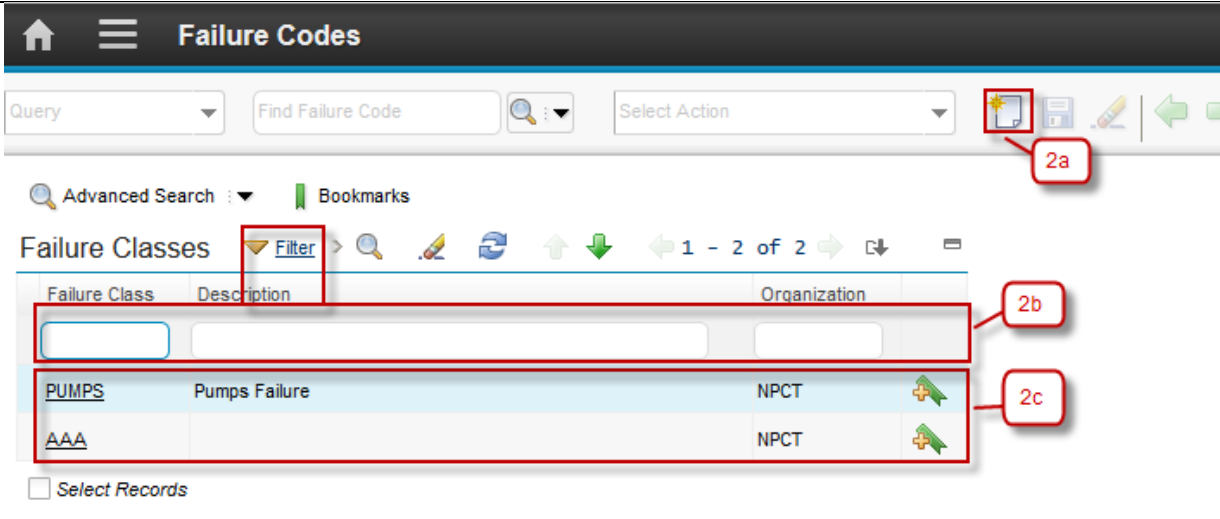
For every corrective work, whether it is CM or SR work (SR work will be discussed next), cause of failure and the solution must be defined before it is being closed. This regulation stipulated that companies have a source of knowledge about the problems that may occur and how to overcome them. A failure codes should be defined for each work order, containing:

- *Problem.* Define the problem that occurs.
- *Cause.* Define the cause of the problem.
- *Remedy.* Define the solution of the problem.

A failure codes must be defined by *Service Engineer* for CM/SR work order before the work order is submitted for closing. Failure codes can be chosen from available list, defined in **Failure Codes** application. Below are details about Failure Codes applications accessibility:

1


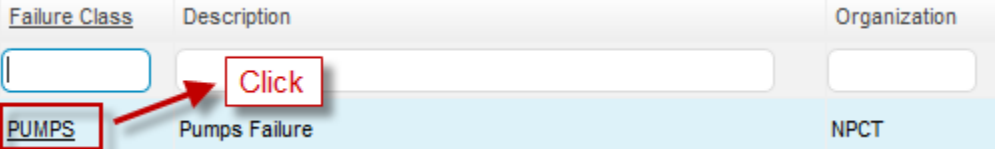
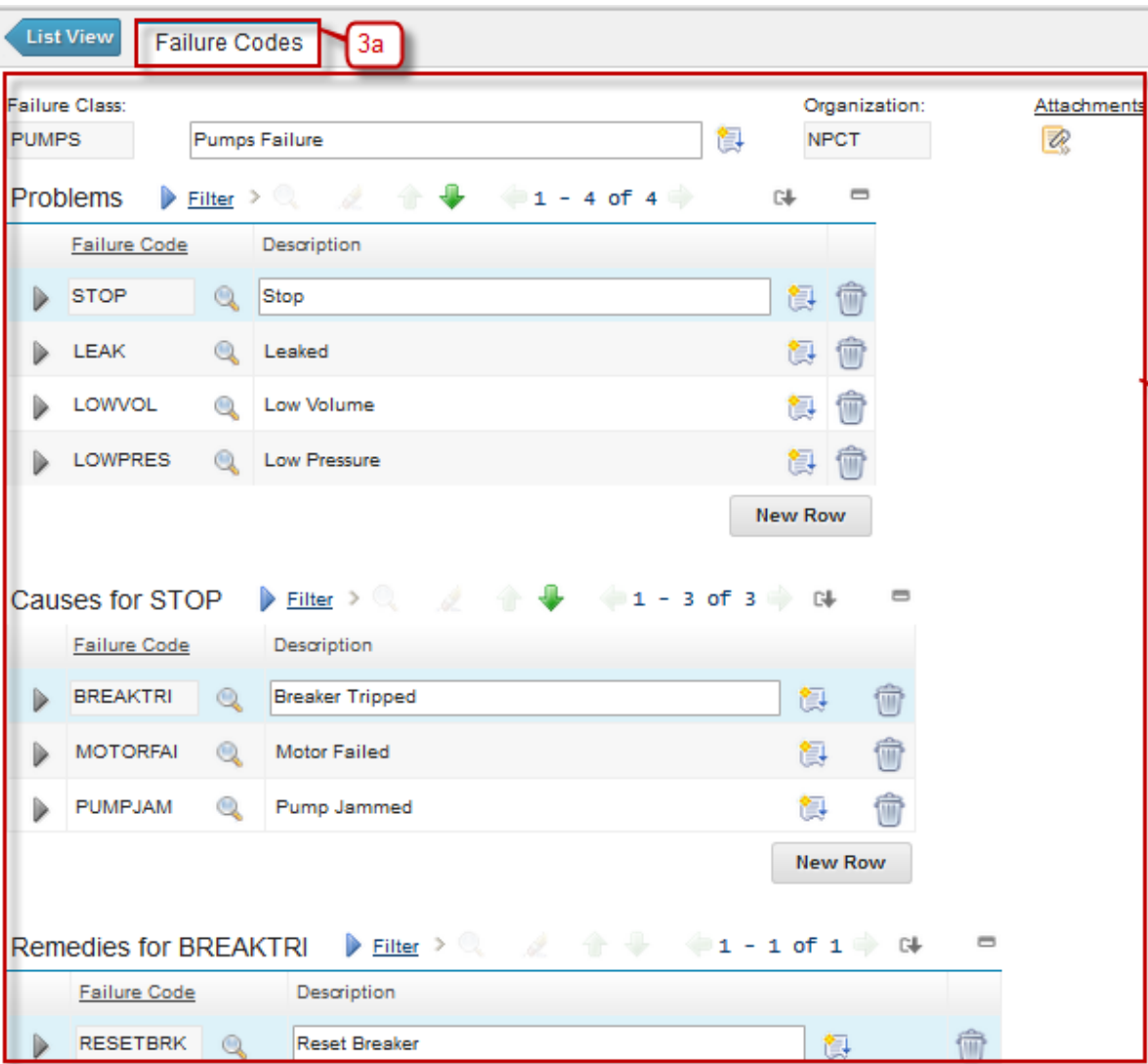
From **Go To** menu, choose **Assets – Failure Codes**

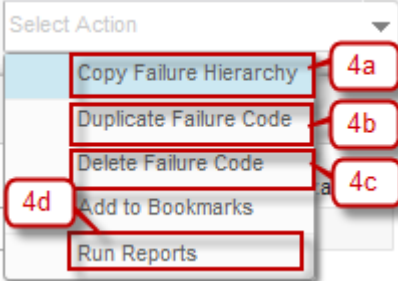
2


Header Bar will give information that user is in the Failure Codes application.

2.a **New Failure Code** button, to add new Failure Code

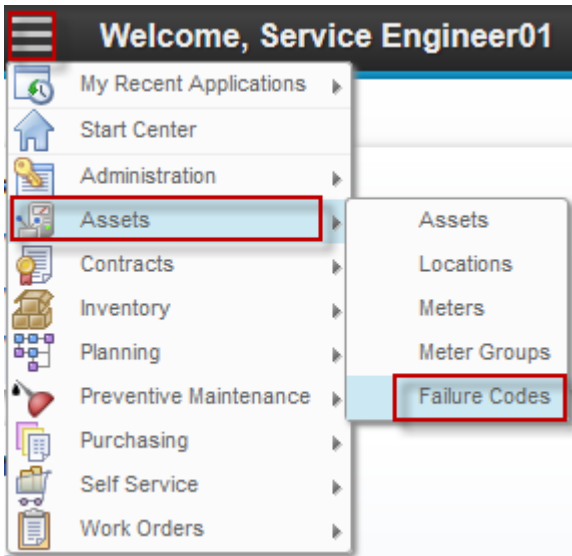
2.b **Filter** Failure Classes by **Failure Class, Description, and Organization**.

	2.c List of all failure class based on filtering result.
3	Choose one Failure Class, to show the details of the Failure Class.  Will show display the details, see the figure below.  3.a Failure Codes tab, to view and update details of Failure Code 3.b Detail information of the Problem, Causes, and Remedies.
4	User can perform some action related to the Failure Code from Select Action menu (on top of the form).

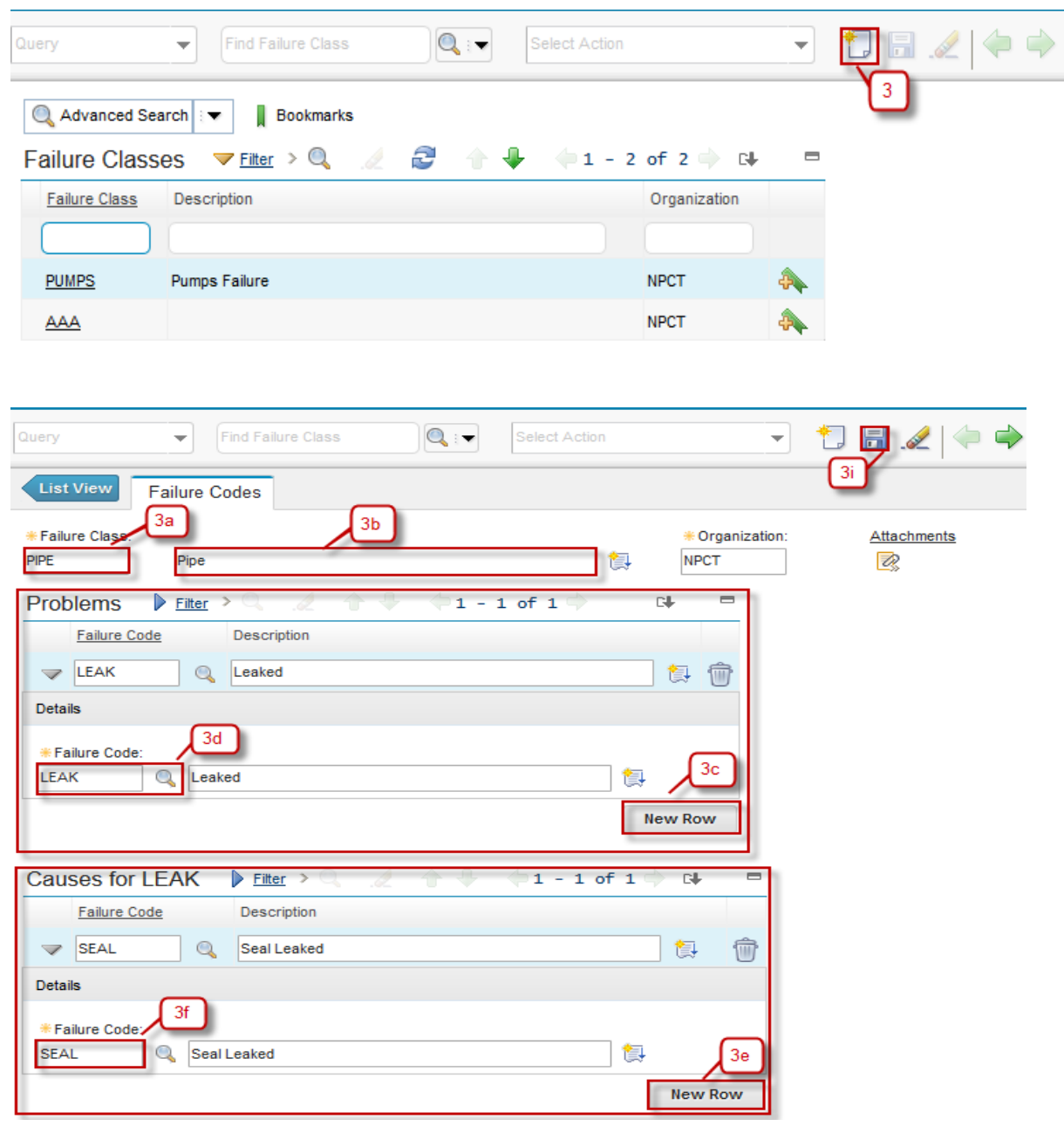
	4.a Copy Failure Hierarchy action to copy failure hierarchy of every problem. 4.b Duplicate Failure Code, to create new failure code by duplicating most value of the current failure code. 4.c Delete Failure Code, to delete failure code. 4.d Run Reports, to run reports of failure code.
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5.2.5.1 Create Failure Codes

When new failure code need to be defined, or new problem, cause, remedy need to be added in a failure code, *Service Engineer* can perform that in **Failure Codes** application, by following below steps:

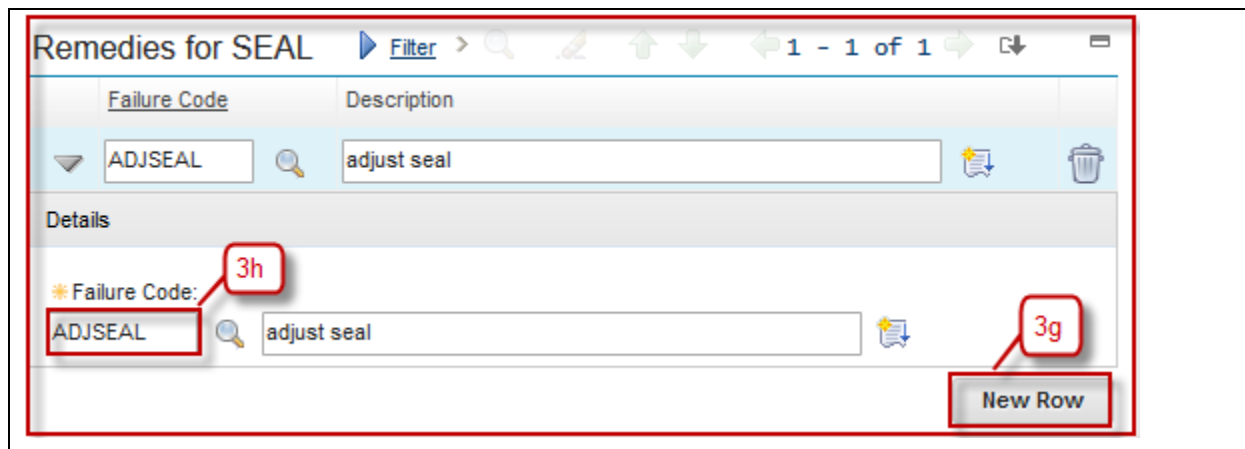
Step	Action
1. Login as Service Engineer. User name : se01, password : password. 2. Go to Work Order Application by click  (GoTo) – Assets – Failure Codes	
3. Click New Work Order  button 3.a Input the failure class name on Failure Class column 3.b Input description of that failure class. 3.c Click New Row on Problems list.	

- 3.d Select Failure Code of Problem.
- 3.e Click **New Row** on **Causes** list.
- 3.f Select Failure Code of Causes.
- 3.g Click **New Row** on **Remedies** list.
- 3.h Select Failure Code of Remedies.
- 3.i Click  **Save** button.



The screenshot displays the Agile Work Management interface, showing the process of adding a new failure code to a problem and its causes. The interface is divided into several sections:

- Top Bar:** Contains a "Query" dropdown, a "Find Failure Class" search bar, a "Select Action" dropdown, and a "Save" button (labeled 3).
- Failure Classes Table:** A table with columns "Failure Class", "Description", and "Organization". It shows two rows: "PUMPS" (Pumps Failure, NPCT) and "AAA" (NPCT). A "New Row" button is visible at the bottom right of the table.
- Failure Codes Section:** A section titled "Failure Codes" with a "List View" button. It contains a "Failure Class" dropdown (labeled 3a) set to "PIPE", a "Description" field (labeled 3b) containing "Pipe", and an "Organization" dropdown (labeled 3i) set to "NPCT".
- Problems Section:** A section titled "Problems" with a "Filter" button. It shows a "Failure Code" dropdown (labeled 3d) set to "LEAK" and a "Description" field (labeled 3c) containing "Leaked". A "New Row" button is visible at the bottom right of the section.
- Causes for LEAK Section:** A section titled "Causes for LEAK" with a "Filter" button. It shows a "Failure Code" dropdown (labeled 3f) set to "SEAL" and a "Description" field (labeled 3e) containing "Seal Leaked". A "New Row" button is visible at the bottom right of the section.



Remedies for SEAL

Filter > 1 - 1 of 1

Failure Code	Description
ADJSEAL	adjust seal

Details

Failure Code: ADJSEAL

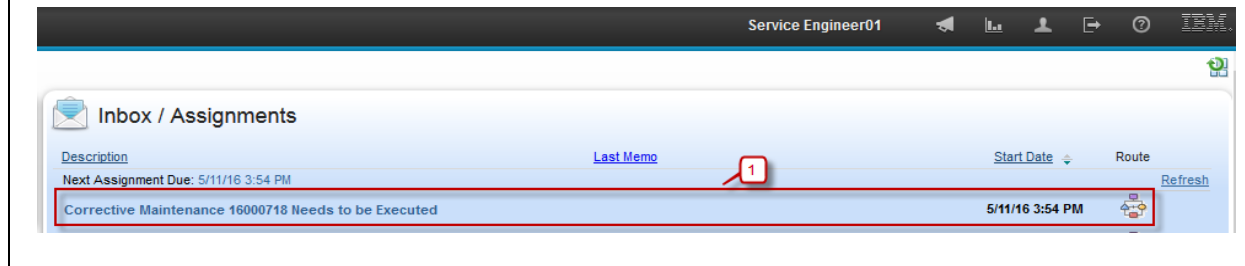
Description: adjust seal

New Row

5.2.6 CM Work Completion

When CM work is finished, *Service Engineer* should input the work result in the CM work order (including failure codes). After that, the work order must be submitted for closing. Below are step by step to complete SM Work Order:

Step	Action
1.	Login as Service Engineer. Username : se01, password : password. Choose one of related work order which has "INPRG" status and work type 'CM' (Corrective Maintenance).
2.	Select the Failure Code for this Work Order, click » button to select failure code and then click Save button.
3.	Then click  (route workflow) and the Route Workflow button appearance returned back into  , indicates that the workflow is started.
4.	Choose Complete Work Order .
5.	Input Memo (if needed).
6.	Then click OK .
7.	Dialog box Report Downtime will be appeared and field Start Date and End Date automatically filled. Value of Hours is interval between Start Date and End Date.
8.	Click OK
9.	Status change to 'COMP (Complete).



Service Engineer01

Inbox / Assignments

Description	Last Memo	Start Date	Route
Corrective Maintenance 16000718 Needs to be Executed		5/11/16 3:54 PM	Route

List View | **Work Order** | Plans | Assignments | Related Records | Actuals | Safety Plan | Log | Failure Reporting

Work Order: 16000718 | Gantry Crane Corrective Maintenance | Site: NPCT1

Location: NP1BERTH >> NPCT1 Berth Location | Work Type: CM

Asset: NP100106 >> Gantry Crane | GL Account: 281-4005-1-01

Class: WORKORDER | Storeroom Material Status: >> | Status: INPRG

Failure Class: **PIPE** >> | Work Package Material Status: PARTIAL >> | Status Date: 5/11/16 4:13 PM

Problem Code: | Material Status Last Updated: 5/11/16 3:45 PM | Inherit Status Changes? ☒

Parent WO: | Closing Financial Period: | Is Task? ☐

Under Flow Control? ☐

Suspend Flow Control? ☐

Query | Find Work Order | Select Action | 

List View | **Work Order** | Plans | Assignments | Related Records | Actuals | Safety Plan | Log | **Failure Reporting**

Work Order: 16000718 | Gantry Crane Corrective Maintenance | Site: NPCT1

Attachments

Complete Workflow Assignment

Task: Corrective Maintenance 16000718 Needs to be Completed

Action: **Complete Work Order**

Memo:

Earlier Memos | Filter > | 0 - 0 of 0

Memo	Person	Transaction Date
There are no rows to display.		

OK | Reassign | Cancel

Report Downtime

To change the asset's current status from UP to DOWN or DOWN to UP, select Change Status. To record the start and end of the asset's downtime, which leaves the asset in its current status, select Report Downtime.

Asset:
NP100106 Gantry Crane

Asset Up?
☒

Downtime Report

☐ Change Status
☒ Report Downtime

Start Date:
5/11/16 2:42 PM

End Date:
5/11/16 5:23 PM

Hours:
2:41

Downtime Code:

Start Date Default: Reported Date
Downtime Type: Operational

OK Cancel

List View Work Order Plans Assignments Related Records Actuals Safety Plan Log Failure Reporting

Work Order:
16000718 Gantry Crane Corrective Maintenance

Location:
NP1BERTH NPCT1 Berth Location

Asset:
NP100106 Gantry Crane

Class:
WORKORDER

Site:
NPCT1

Work Type:
CM

GL Account:
281-4005-1-01

Storeroom Material Status:

Attachments

Status:
COMP

Status Date:
5/11/16 5:23 PM

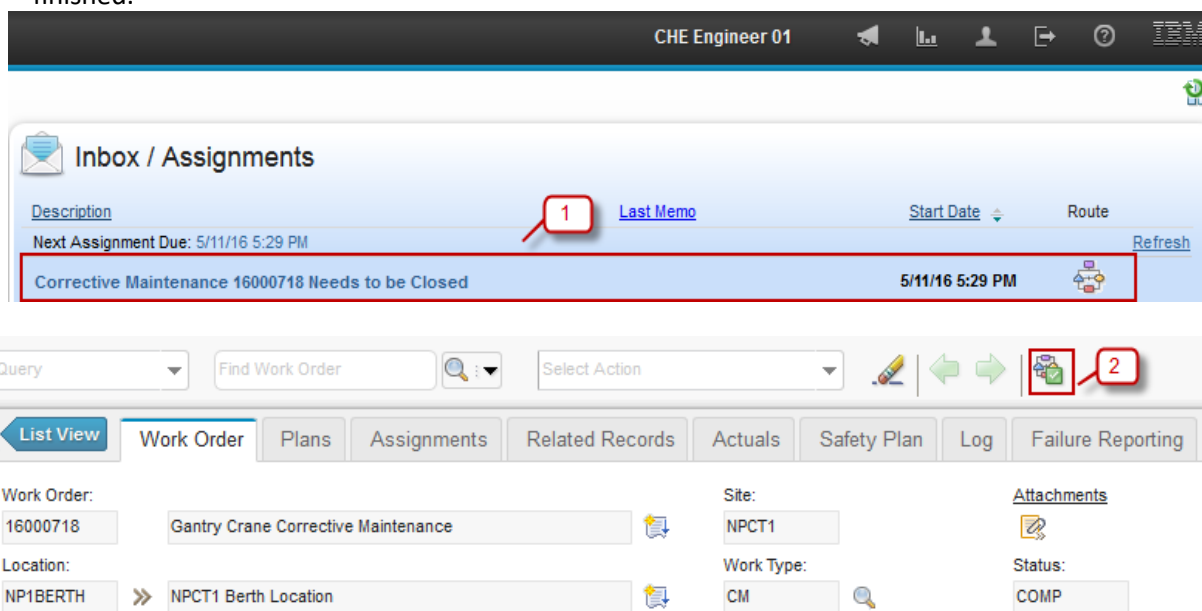
Inherit Status Changes?
☒

5.2.7 CM Work Closing Approval by CHE Engineer

Below are steps to approve a CM work closing by *CHE Engineer*. For rejection, revision request, and revision will not be discussed. They can refer to rejection, revision request, and revision of **SM Work Cycle** in step point **1.2.5 SM Request Revision Work Order** and **1.2.6 SM Revise and Re-submit Work Order** training material.

As a note, request revision for CM work closing will direct the assignment back to *Service Engineer*. After revision is done, assignment will be directed back to *CHE Engineer* for closing approval.

Step	Action
1.	Login as CHE Engineer. Username : che01, password: password. On Start Center Operation an assignment to close the Work Order will appears in Inbox/Assignment portlet, click Description of the assignment to go to related Work Order
2.	Then click  (route workflow) and dialog box Complete Workflow Assignment will appears.
3.	Choose Close Work Order
4.	Input Memo (if needed)
5.	Then click OK
6.	Status changed to ' CLOSE ' (Close).
7.	The Route Workflow button appearance returned back into  , indicates that the workflow is finished.



Complete Workflow Assignment

Task:
Corrective Maintenance 16000718 Needs to be Closed

Action:

☒ Close Work Order 3

☐ Revise Work Order

Memo:

4

Earlier Memos [Filter](#) > 0 - 0 of 0

Memo	Person	Transaction Date
There are no rows to display.		

5

Query Find Work Order Select Action

7

[List View](#) [Work Order](#) [Plans](#) [Assignments](#) [Related Records](#) [Actuals](#) [Safety Plan](#) [Log](#) [Failure Reporting](#)

Work Order:
16000718 Gantry Crane Corrective Maintenance

Location:
NP1BERTH >> NPCT1 Berth Location

Site:
NPCT1

Work Type:
CM

Attachments

Status: 6

5.3 Exercises

6 Service Recovery (SR) Work

6.1 SR Work Overview

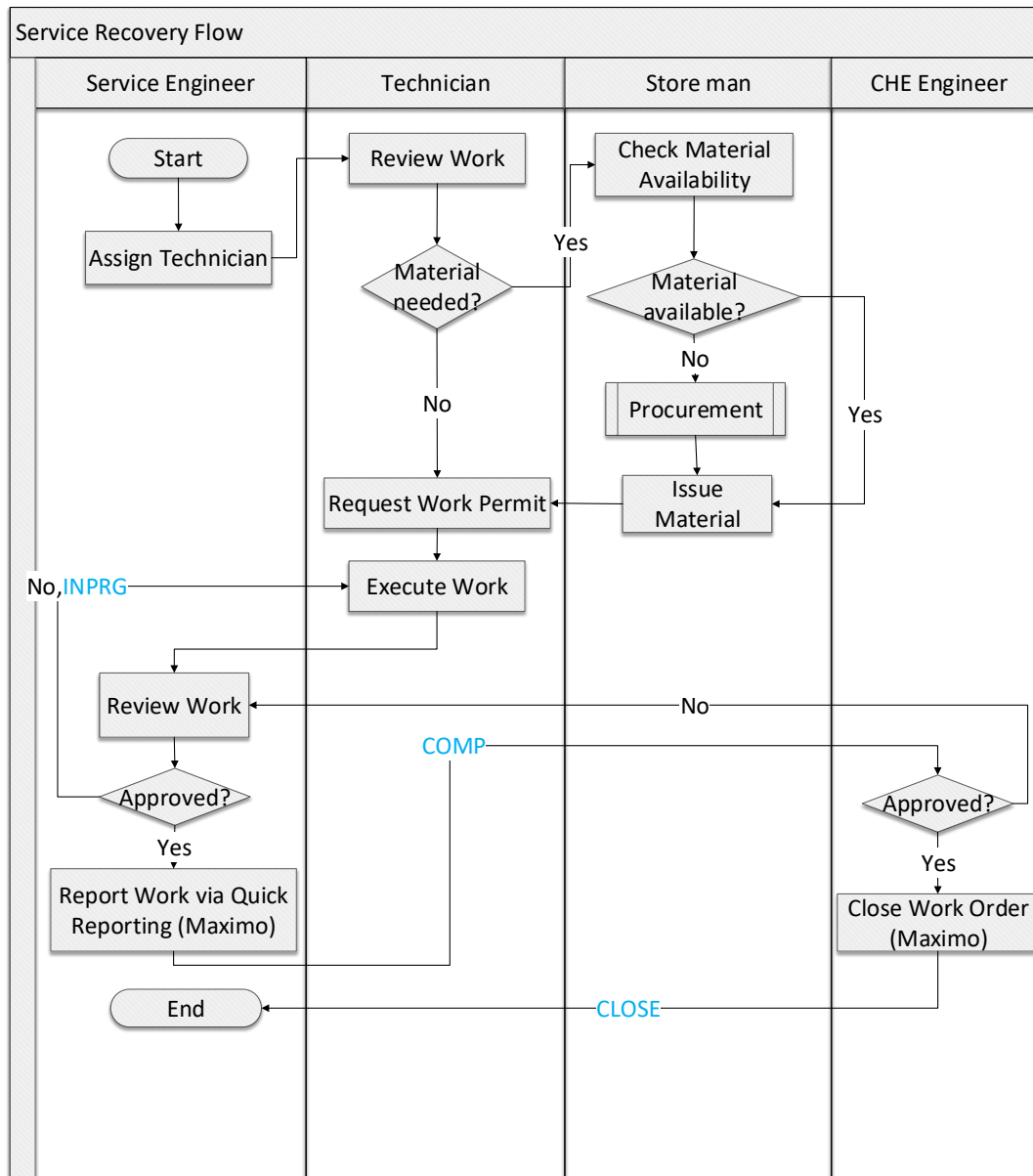
Service Recovery (SR) work is a corrective maintenance work with high level of urgency that makes them need to be executed immediately. Because of the high level of urgency, work execution can be done without prior work order. Material usage for this work does not require any prior approval too. After the work finished, report about this work can be registered as SR work order in **Quick Reporting** application.

Information that should be included in a SR work order is as follows:

- Tasks that have been performed
- Any materials or tools used.
- Labor resources used.
- Failure codes containing at least 1 problem, causes, and remedy.

6.2 SR Work Cycle

Below figure show how service recovery will be managed in EMS.



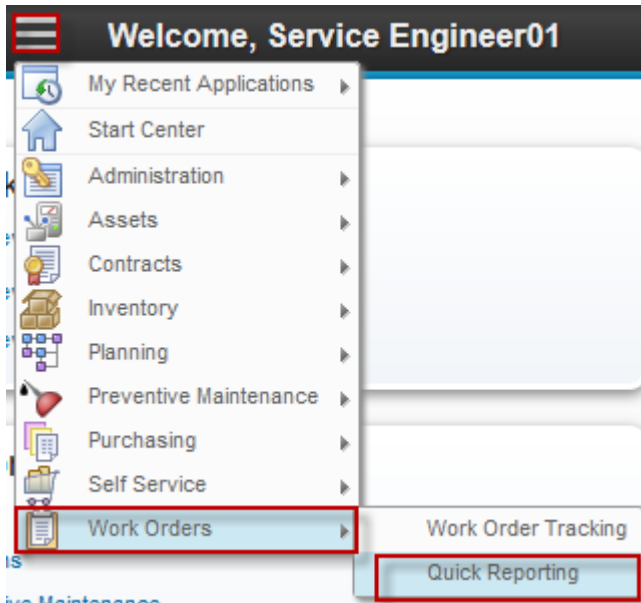
Below are detail scenario for Service Recovery (SR) work order:

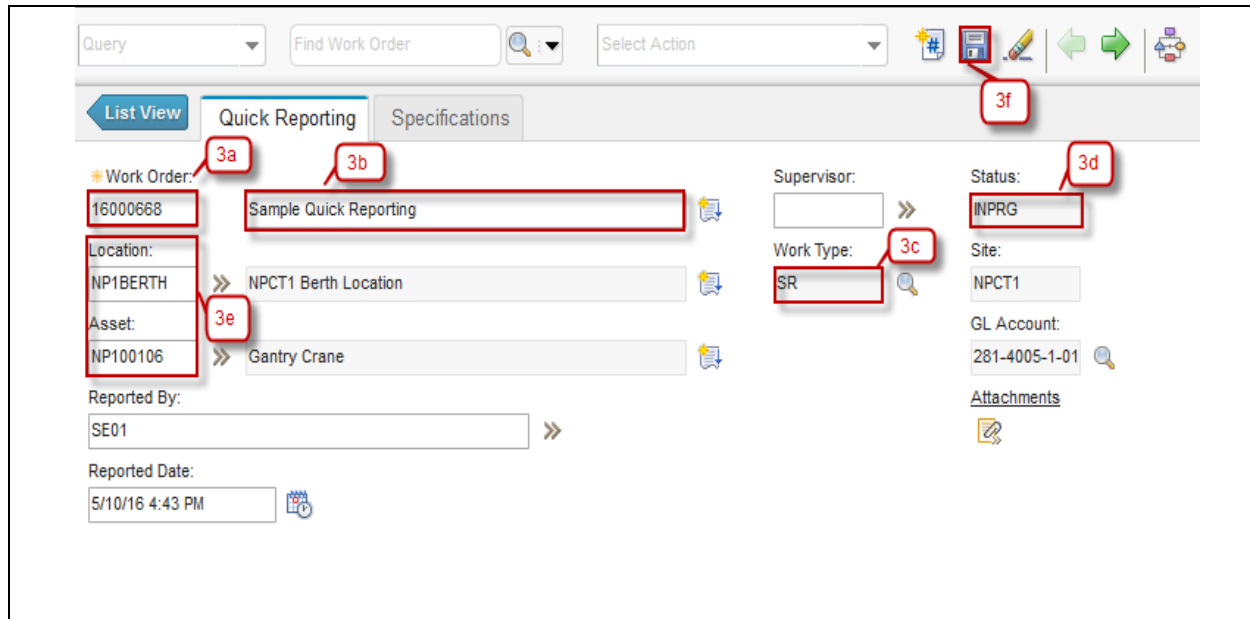
1. SR can be initiated without any prior work order, because it is an emergency. Technician assignment is performed directly. Any material issue from storeroom can be done without any prior approval.
2. After the work is done, *Service Engineer* creates a SR work order as a report of the work in **Quick Reporting** application. *Service Engineer* also include any resources used for this SR.

A closing approval assignment will be sent to *CHE Engineer*, whether the work order can be closed or not.

6.2.1 SR Work Reporting & Submission

After a SR work is finished, Service Engineer should create a work reporting in **Quick Reporting** application. Below are step by step for SR work reporting & submission:

Step	Action
1. Login as Service Engineer. User name : se01, password : password.	
2. Go to Quick Reporting Application by click  (GoTo) – Work Orders – Quick Reporting	
3. Click New Quick Reporting button 	
3a. Work Order number is autonumber.	
3b. Input Work Order Description .	
3c. Default Work Type of Quick Reporting is ' SR ' (Service Recovery).	
3d. Default Status of Quick Reporting is ' INPRG '	
3e. Click  on Asset or Location field and select one value.	
3f. Click  Save .	



4. Add a task by following this steps:

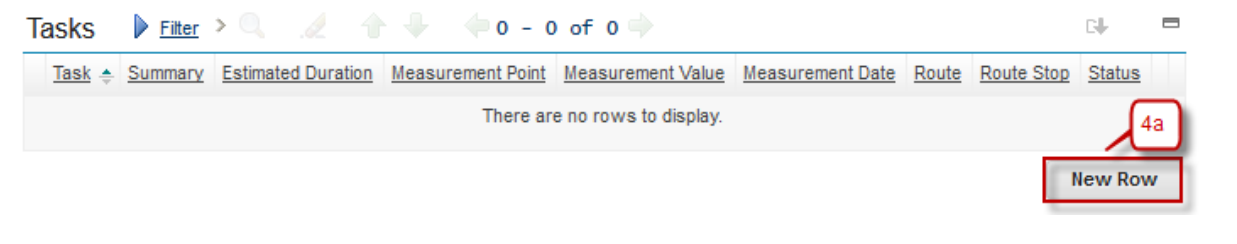
4a. Click **New Row**

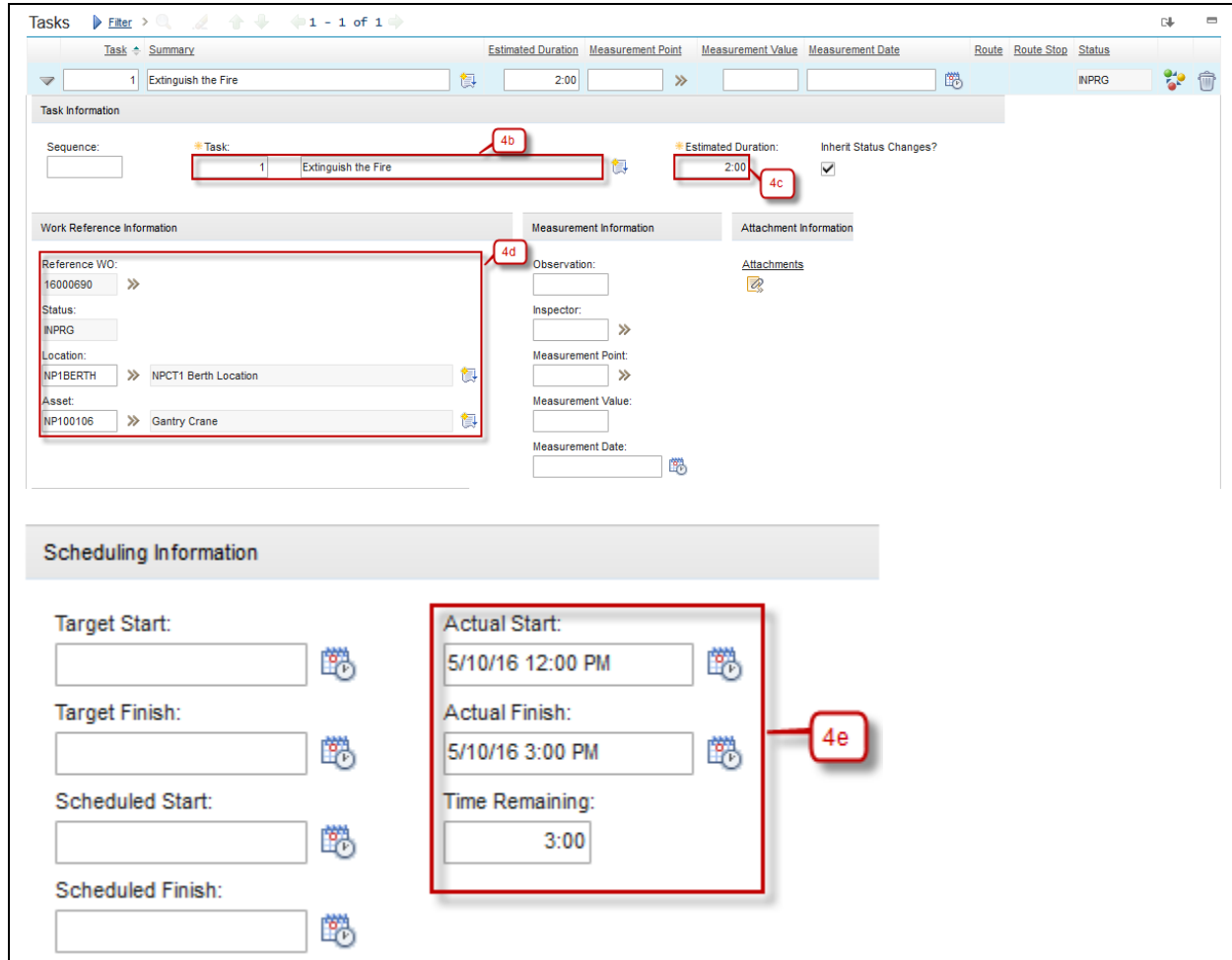
4b. Input Task Description.

4c. Input Estimated Duration.

4d. When inputting task for Work Reference Information Section is automatically filled.

4e. Input the Actual Start and Actual Finish. Value of Time Remaining is Duration minus the number of hours worked so far.





The screenshot displays the 'Tasks' interface with a filter set to '1 - 1 of 1'. The task 'Extinguish the Fire' is selected, showing an estimated duration of 2:00 and a status of INPRG. The interface is divided into several sections:

- Task Information:** Includes fields for Sequence (1), Task (Extinguish the Fire), Estimated Duration (2:00), and Inherit Status Changes? (checked).
- Work Reference Information:** Includes Reference WO (16000690), Status (INPRG), Location (NP1BERTH), and Asset (NP100106).
- Measurement Information:** Includes Observation, Inspector, Measurement Point, Measurement Value, and Measurement Date.
- Attachment Information:** Includes Attachments.
- Scheduling Information:** Includes Target Start, Target Finish, Scheduled Start, and Scheduled Finish. The Actual Start (5/10/16 12:00 PM), Actual Finish (5/10/16 3:00 PM), and Time Remaining (3:00) are also displayed.

Red callouts highlight specific fields: 4b points to the Task field, 4c points to the Estimated Duration field, 4d points to the Reference WO field, and 4e points to the Actual Finish field.

5. Select **Labor** Tab under the page.
- 5a. Click **Select Labor** button.
- 5b. Select associated Labor and click **OK**.
- 5c. After Select Labor, field Labor, Craft, Skill Level, Start Date, Rate, Type is automatically fill. Input End Date when the task finished.
- 5d. Input Start Time.
- 5e. Input End Time.
- 5f. Value of Duration (Hours) is interval between Start Date and End Date.
- 5g. If Labor using from Outside, the Outside Labor Information will automatically fill.

Labor Materials Tools Failure Reporting Meter Readings Log

Labor [Filter](#) >     0 - 0 of 0  

Task	Labor	Name	Approved?	Start Date	Start Time	End Time	Duration (Hours)	Rate
There are no rows to display.								

Enter Time By Crew **Select Labor** Select Planned Labor New Row

Select Labor

Labor [Filter](#) >     1 - 15 of 18  

☐	Labor	Name	Craft	Skill Level	Vendor	Contract	Contract Description
☐	ELECT01	Electrical 01	ELECT				
☐	MECH01	Mechanical 01	MECH				
☐	ELECT02	Electrical 02	ELECT	JUNIOR	MTSUI	1002	
☐	ELECT01	Electrical 01	ELECT	SENIOR	MTSUI	1002	
☐	MECH02	Mechanical 02	MECH	JUNIOR	MTSUI	1002	
☐	MECH01	Mechanical 01	MECH	SENIOR	MTSUI	1002	
☐	ELECT02	Electrical 02	ELECT	JUNIOR	MTSUI	NP11001	NPCT1 Labor Rate Contract
☒	ELECT01	Electrical 01	ELECT	SENIOR	MTSUI	NP11001	NPCT1 Labor Rate Contract
☐	MECH02	Mechanical 02	MECH	JUNIOR	MTSUI	NP11001	NPCT1 Labor Rate Contract
☐	MECH01	Mechanical 01	MECH	SENIOR	MTSUI	NP11001	NPCT1 Labor Rate Contract
☐	ELECT02	Electrical 02	ELECT	JUNIOR	709654	NP1LAB01	Labor Maintenance Contract
☐	ELECT01	Electrical 01	ELECT	SENIOR	709654	NP1LAB01	Labor Maintenance Contract
☐	MECH02	Mechanical 02	MECH	JUNIOR	709654	NP1LAB01	Labor Maintenance Contract

OK Cancel

Labor
Materials
Tools
Failure Reporting
Meter Readings
Log

Labor
Filter
1 - 1 of 1

Task	Labor	Name	Approved?	Start Date	Start Time	End Time	Duration (Hours)	Rate
ELECT01	>>	Electrical 01	☐	5/10/16	6:00 AM	6:00 PM	12:00	175,000.00

Task:
Labor: ELECT01 >> Electrical 01
Approved? ☐

Craft: ELECT >>
Skill Level: SENIOR >>

Start Date: 5/10/16
End Date: 5/10/2016
Type: WORK

Start Time: 6:00 AM
End Time: 6:00 PM

Duration (Hours): 12:00
Line Cost: 2,100,000.00

Rate: 175,000.00

Outside Labor
Premium Pay
Charge Information

Outside? ☒
Vendor: MITSUI >>
Contract: NP11001 >>
Revision: 0

Premium Pay Code:
Premium Pay Hours:
Premium Pay Rate:
Premium Rate Type:

GL Debit Account:
GL Credit Account:
Location: NP1WS >>
Asset: NP100106 >>
Memo:
Recorded as Received:

6. Select **Tools** tab under the page

6a. Click **Select Tools**.

6b. Select tools used for work order and then click **OK**.

6c. Input Quantity, default value of Quantity is 1.00

6d. Input Tool Hours, default value of Tool Hours is 1.00

6e. Input Rate, default value of Rate is 0.00

6f. Value of Line Cost is multiplication result from Quantity and Rate. Click **Save**

Agile – Proprietary/Confidential

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Labor Materials **Tools** Failure Reporting Meter Readings Log

Tools [Filter](#) >     0 - 0 of 0  

Task	Tool	Description	Quantity	Tool Hours	Rate	Line Cost	Outside?	Location
There are no rows to display.								

6a 

Select Tools Select Planned Tools Select Issued Tools New Row

Select Tools

Tools [Filter](#) >     1 - 10 of 10  

☐	Tool	Description	Outside?
☐	TOOL001	Fluke Multimeter	☐
☒	TOOL002	Hammer	☐
☐	TOOL03	Saw	☐
☐	NP1000016		☐
☐	NP1000017	a	☐
☐	NP1000035		☐
☐	NP1000054	Fluke Multimeter 20A, 1000VDC, 440VAC	☐
☐	NP1000056	Ttest Tool	☐
☐	NP1000057	Np2 the poci d sss	☐
☐	NP1000087	Fluke Multimeter 20A, 1000VDC, 440VAC	☐

6b 

OK Cancel

Labor Materials **Tools** Failure Reporting Meter Readings Log

Tools Filter > 1 - 1 of 1

Task	Tool	Description	Quantity	Tool Hours	Rate	Line Cost	Outside?	Location
	TOOL002	Hammer	1	1:00	0.00	0.00	☐	NP1WS

Details

Task:

Tool: TOOL002 Hammer

Outside? ☐

Location: NP1WS NPCT1 Workshop Repair Facility

Asset: NP100106 Gantry Crane

Rotating Asset:

Quantity: 1 (6c)

Tool Hours: 1:00 (6d)

Rate: 0.00 (6e)

Line Cost: 0.00 (6f)

GL Debit Account:

GL Credit Account:

Entered By: SE01

Entered Date: 5/10/16 10:42 AM

Select Tools Select Planned Tools Select Issued Tools New Row

7. Select Failure Reporting tab under the page.

7a. Select Failure Class.

7b. Click Select Failure Codes, dialog box will be appeared and then select failure codes.

Labor Materials Tools **Failure Reporting** Meter Readings Log

Failure Class: PUMPS (7a) Pumps Failure

Failed Date: 5/10/16 10:51 AM

Remarks:

Remark Date:

Failure Codes Filter > 1 - 3 of 3

Type	Failure Code	Description
PROBLEM	LEAK	Leaked
CAUSE	SEAL	Seal Leaked
REMEDY	RPLCSEAL	Replace Seal

7b Select Failure Codes

8. Complete Quick Reporting

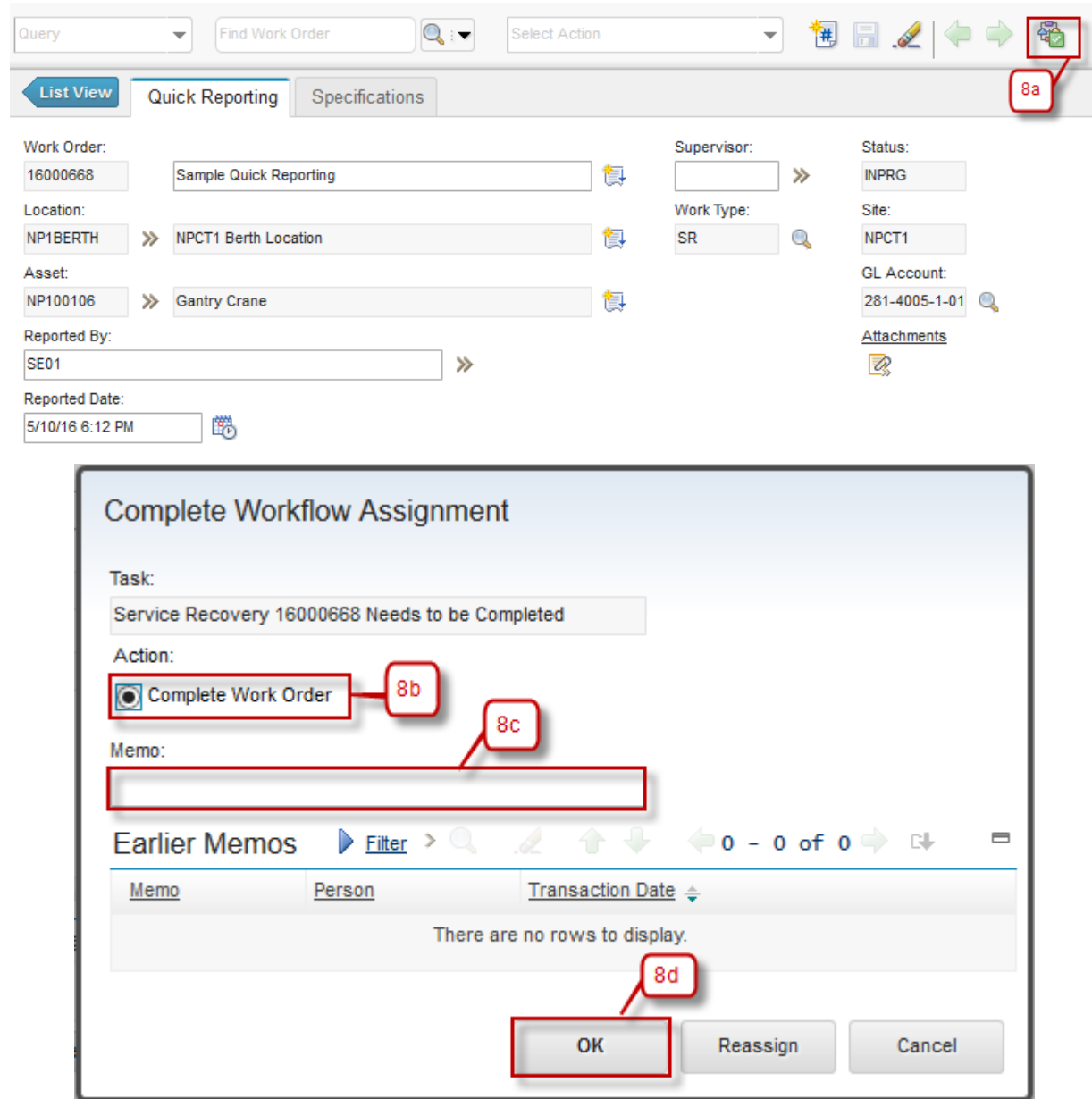
8a. Click  (Route Workflow) button.

8b. Dialog box **Complete Workflow Assignment** will appear. Choose action **Complete Work Order**.

8c. Input Memo (if needed).

8d. Click **OK**.

8e. Status will be changed to '**COMP**' (Complete).



Query Find Work Order Select Action

List View Quick Reporting Specifications

Work Order: 16000668 Sample Quick Reporting

Location: NP1BERTH NPCT1 Berth Location

Asset: NP100106 Gantry Crane

Reported By: SE01

Reported Date: 5/10/16 6:12 PM

Supervisor: Status: INPRG

Work Type: SR Site: NPCT1

GL Account: 281-4005-1-01

Attachments

Complete Workflow Assignment

Task: Service Recovery 16000668 Needs to be Completed

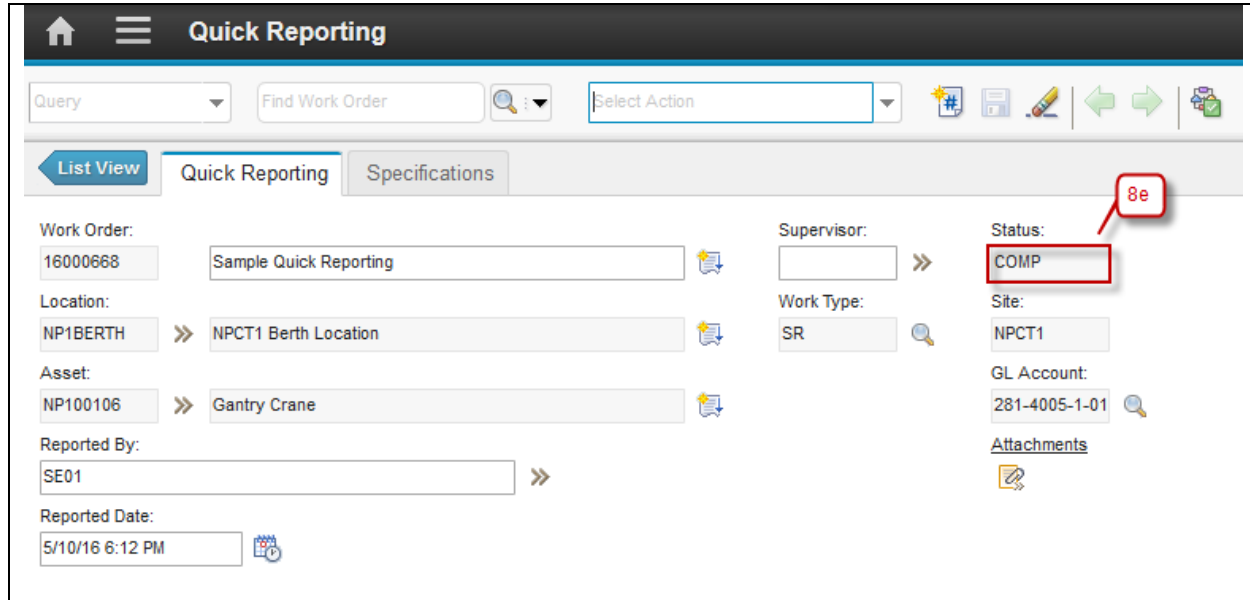
Action: ☒ Complete Work Order

Memo:

Earlier Memos Filter 0 - 0 of 0

Memo	Person	Transaction Date
There are no rows to display.		

OK Reassign Cancel



6.2.2 SR Work Closing Approval by CHE Engineer

Below are steps to approve a SR work closing by *CHE Engineer*. For revision request, and revision will not be discussed. They can refer to revision request, and revision of **SM Work Cycle** in step point **1.2.5 SM Request Revision Work Order** and **1.2.6 SM Revise and Re-submit Work Order** training material.

As a note, request revision for SR work closing will direct the assignment back to *Service Engineer*. After revision is done, assignment will be directed back to *CHE Engineer* for closing approval.

Step	Action
1.	Login as CHE Engineer. Username : che01, password: password. On Start Center Operation An assignment to close SR Work Order will appears in Inbox/Assignment portlet, click Description of the assignment to go to related Work Order
2.	Click  (Route Workflow) button and dialog box Complete Workflow Assignment will appears.
3.	Choose Close Work Order .
4.	Input Memo (if needed).
5.	Then click OK .
6.	Status changed to 'CLOSE'.
7.	The Route Workflow button appearance returned back into  , indicates that the workflow is finished.

CHE Engineer 01

Inbox / Assignments

Description	Last Memo	Start Date	Route
Next Assignment Due: 5/11/16 1:55 PM Refresh			
Corrective Maintenance 16000717 Needs to be Approved	1	5/11/16 1:55 PM	
Service Recovery 16000668 Needs to be Closed		5/11/16 1:40 PM	

Query Find Work Order Select Action      

List View Quick Reporting Specifications

Work Order: 16000668 Sample Quick Reporting

Location: NP1BERTH NPCT1 Berth Location

Asset: NP100106 Gantry Crane

Reported By: SE01

Reported Date: 5/10/16 6:12 PM

Supervisor: Status: INPRG

Work Type: SR Site: NPCT1

GL Account: 281-4005-1-01

[Attachments](#)

Complete Workflow Assignment

Task: Service Recovery 16000668 Needs to be Closed

Action: ☒ Close Work Order 3 ☐ Revise Work Order

Memo: 4

Earlier Memos [Filter](#) 0 - 0 of 0

Memo	Person	Transaction Date
There are no rows to display.		

5 OK Reassign Cancel

Query Find Work Order  Select Action    

List Quick Reporting Specifications

Work Order: 16000668 Sample Quick Reporting 

Location: NP1BERTH >> NPCT1 Berth Location 

Asset: NP100106 >> Gantry Crane 

Reported By: SE01 >>

Reported Date: 5/10/16 6:12 PM 

Supervisor: >>

Work Type: SR 

Status: CLOSE 

Site: NPCT1

GL Account: 281-4005-1-01 

[Attachments](#)

7

6

6.3 Exercises